# FOOD PROCESSING TECHNOLOGY

This course provides comprehensive knowledge of food science, preservation, safety, and industrial processes, covering topics from raw material preparation to packaging. Key areas include thermal processing, fermentation, quality control, and sustainable technologies to ensure food safety and quality for industrial applications.

## Core Modules and Topics

- **Module 1: Introduction to Food Science & Technology**
  - Scope and trends in the food industry
  - Basic food constituents: carbohydrates, proteins, fats, water, vitamins, and minerals.

### • Module 2: Raw Material Handling & Preparation

- Cleaning, sorting, grading, and peeling techniques.

### • Module 3: Food Preservation Techniques

- Low-temperature preservation (cooling, Refrigeration, Freezing)
- Thermal processing (Pasteurization, Sterilization, Blanching, UHT).
- Drying and dehydration technologies
- Fermentation and chemical preservatives.

### • Module 4: Specialized Food Processing Technologies

- Cereal processing (milling, baking)
- Fruits and vegetable processing
- Animal products processing (meat, dairy, fish)
- Fats and oils technology

### • Module 5: Food safety, Quality Control, and Management

- HACCP (Hazard Analysis and Critical Control Points) and food safety systems
- Quality assurance, hygiene, and sanitation in processing units
- Sensory evaluation and food analysis techniques

- ## Module 6: Packaging and storage
    - Packaging materials and technologies
    - Shelf-life testing and storage regulations

- ## Module 7: Advanced and Emerging Technologies
    - High-pressure processing, Ohmic heating, pulse electric field processing.
    - Sustainable food processing

## Practical & Project Component

- **Laboratory sessions:** Hands-on experience with unit operations, testing, and processing equipment.
- **Industrial Training/Internships:** In-plant training for practical skills in industrial settings
- **Product Development:** Project work on creating new food products.

## Key Learning Outcomes

- Understand and apply principles of food chemistry, microbiology, and engineering
- Operate food processing machinery and apply preservation techniques
- Implement safety and quality standards (HACCP)
- Manage food processing waste and ensure sustainability.

## Typical Training Format

- **Introductory:** 5-day workshops covering basics
- **Comprehensive:** Module based training
- **Methods:** Classroom lessons paired with participatory demonstrations/practical sessions